Linear Regression

Machine Learning 1 — Lecture 2

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Robert Peharz
Institute of Machine Learning and Neural Computation
Graz University of Technology

Regression: Predict a *continuous* **target variable** from one or more **input variables**. Hence, a form of **supervised learning.**

input variables (also **independent variables**, **features**):

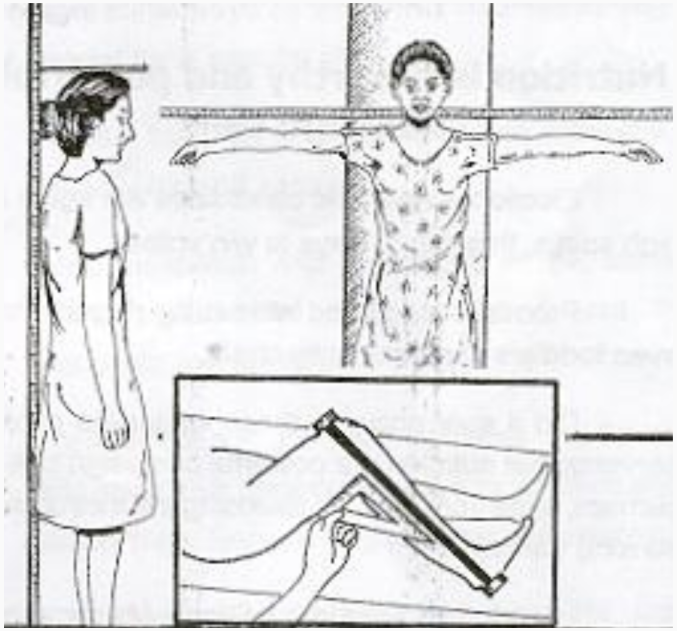
$x_1, x_2, \ldots, x_D \in \mathbb{R}$, collected in a vector $\boldsymbol{x} = (x_1, x_2, \ldots, x_D)^T \in \mathbb{R}^D$

target variable (also **dependent variable**, **output variable**, or **ground truth**):

$$y \in \mathbb{R}$$

Height Regression

Example



knee height [cm]	arm span [cm]	body height [cm]
57	181	185
55	177	177
53	167	176
...	...	...

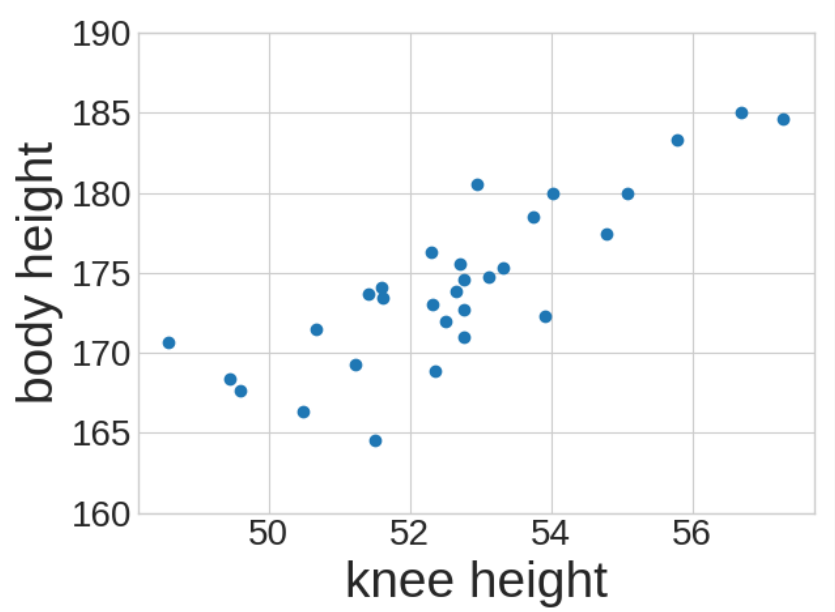
- you need to measure the body height of a hospitalized patient, who is bed tied and cannot stand to take a measurement
- say you have collected data about the knee length, arm span and height of N persons
- you might now learn to predict patients' height from their arm span and knee height, thus "indirectly measure" the height of the hospitalized patient

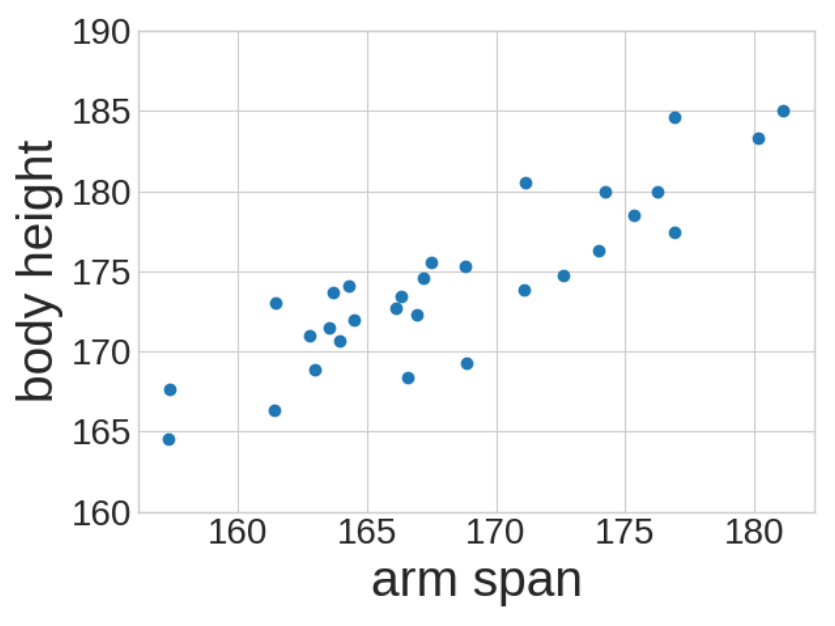
Height Regression – Data

Example

knee height [cm]	arm span [cm]	height [cm]
57	181	185
55	177	177
53	167	176
...	...	...

$N = 30$ data points



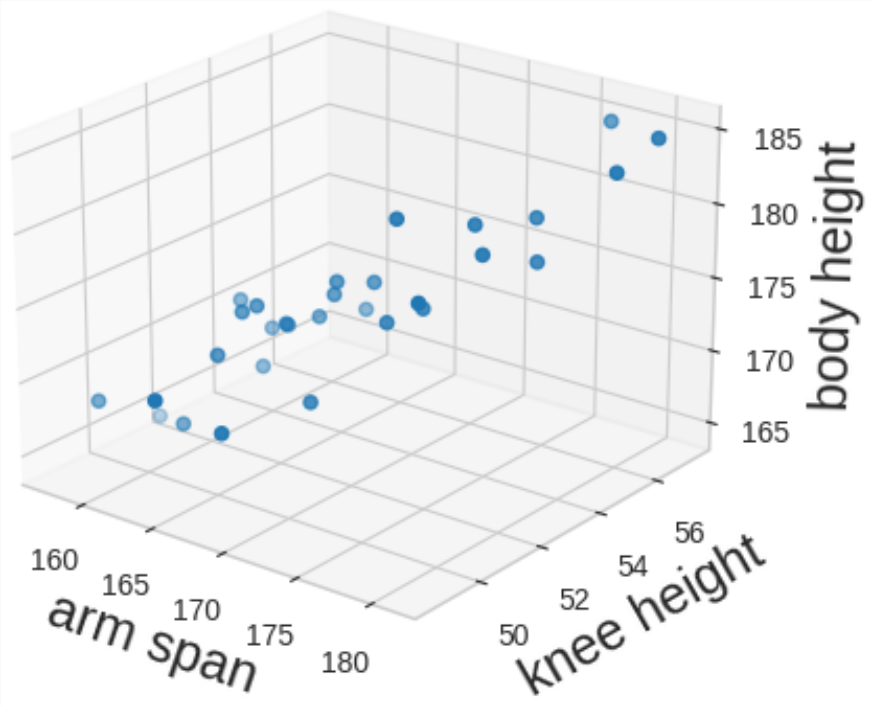


Height Regression – Data

Example

knee height [cm]	arm span [cm]	height [cm]
57	181	185
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53	167	176
...	...	...

$N = 30$ data points



Regression

The target y is a **"noisy" function** of the input variables $\boldsymbol{x}$:

$$y = f(\boldsymbol{x}) + e \qquad\qquad e\text{: } \textbf{noise term}$$

Assume we have a **training data set** with N examples:

$$\mathcal{D} = \left\{(\boldsymbol{x}^{(1)}, y^{(1)}), (\boldsymbol{x}^{(2)}, y^{(2)}), \ldots, (\boldsymbol{x}^{(N)}, y^{(N)})\right\}$$

$\boldsymbol{x}^{(i)} \in \mathbb{R}^D$, $\boldsymbol{x}^{(i)} = (x_1^{(i)}, \ldots, x_D^{(i)})^T$: D-dimensional vector
$y^{(i)} \in \mathbb{R}$: target value
$(\boldsymbol{x}^{(i)}, y^{(i)})$: i^{th} **training example** (input-output pair)

Goal: Learn the function f from $\mathcal{D}$.

Searching over **all possible** functions $f : \mathbb{R}^D \mapsto \mathbb{R}$ is challenging.

Thus, one usually considers a **restricted** class of functions, the so-called **hypothesis space** $\mathcal{H}$ (also called **model class**, **function space**, etc.).

Often, the hypothesis space is given via some **parametrized function**

$$f_{\boldsymbol{\theta}}(\boldsymbol{x}) = f(\boldsymbol{x}, \boldsymbol{\theta})$$

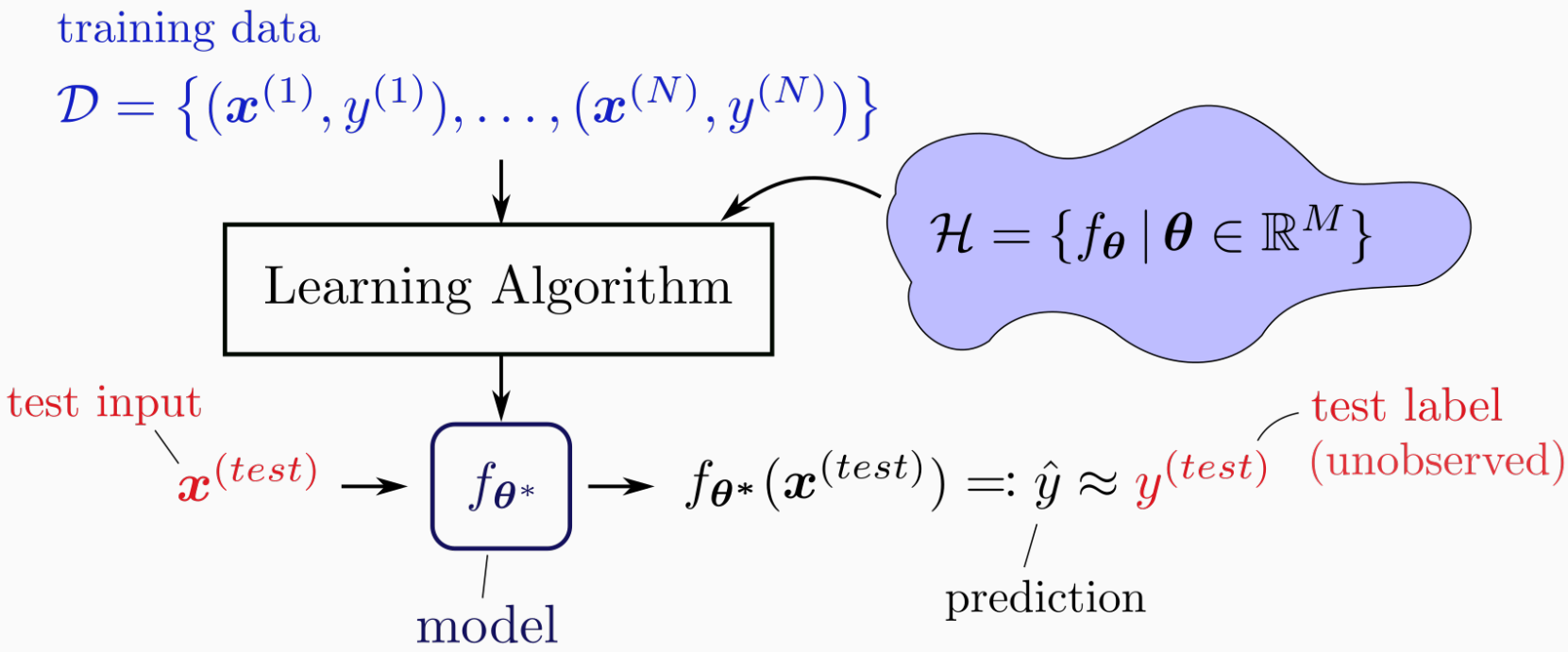
where $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_M)^T$ is an M-dimensional **parameter vector**.

Each $\boldsymbol{\theta} \in \mathbb{R}^M$ describes some function. Formally, the hypothesis space is the **set of functions**

$$\mathcal{H} = \{f_{\boldsymbol{\theta}} \mid \boldsymbol{\theta} \in \mathbb{R}^M\}$$

where $\boldsymbol{\theta}$ ranges over all possible M-dimensional vectors.

The Overall Picture



(Same picture generally applies to **supervised learning**, not only regression)

When the hypothesis space is restricted to all **affine functions**

$$\mathcal{H} = \left\{ f_{\boldsymbol{\theta}}(\boldsymbol{x}) := \boldsymbol{w}^T \boldsymbol{x} + b \,|\, \boldsymbol{w} \in \mathbb{R}^D, b \in \mathbb{R} \right\}$$

we speak of **linear regression**.* Thus, each function is of the form

$$f_{\boldsymbol{\theta}}(\boldsymbol{x}) = \boldsymbol{w}^T \boldsymbol{x} + b = \left(\sum_{d=1}^{D} w_d x_d \right) + b$$

The parameter vector $\boldsymbol{\theta}$ consists of **bias** b and **weight vector** $\boldsymbol{w} = (w_1, w_2, \ldots, w_D)^T$:

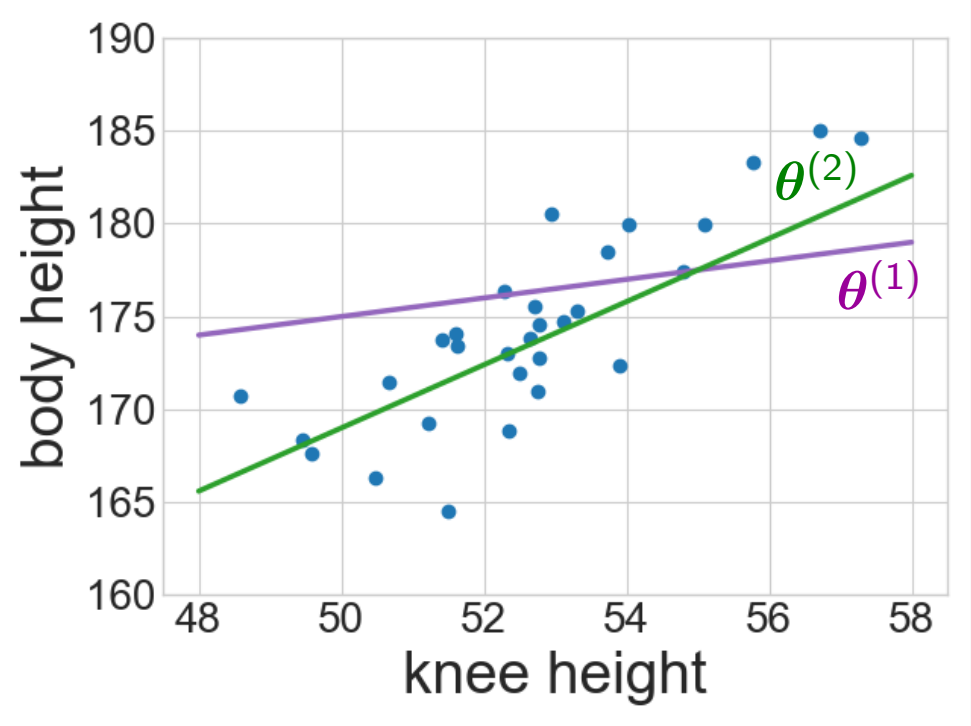
$$\boldsymbol{\theta} = (b, w_1, w_2, \ldots, w_D)^T \in \mathbb{R}^{D+1} \qquad \text{(i.e. } M = D + 1\text{)}$$

*The distinction between *affine* and *linear* function is sloppy at times.

How to pick suitable parameters?

Example

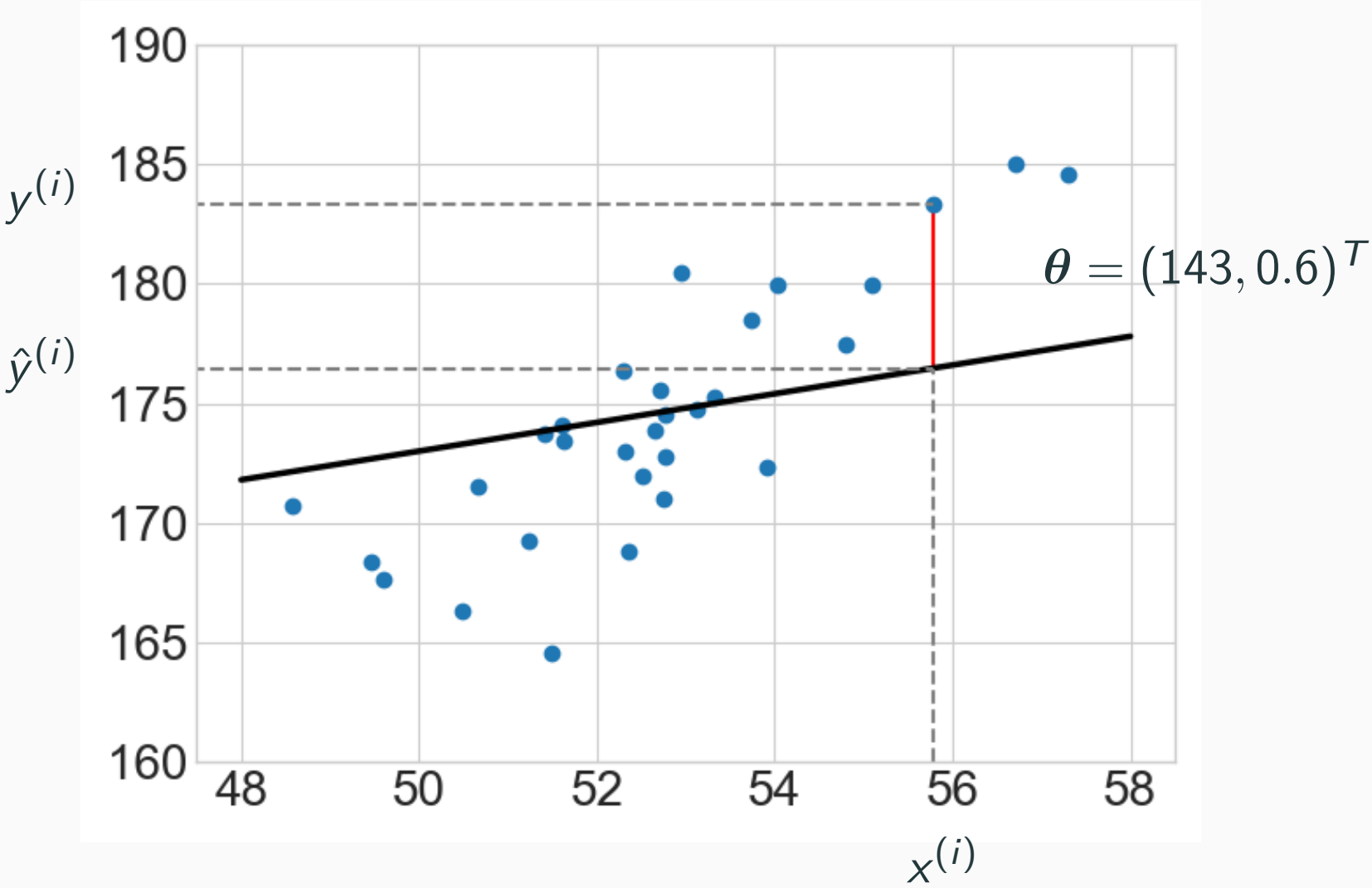
knee height [cm]	height [cm]
57	185
55	177
53	176
...	...



- predict height from knee height (x, 1 input)
- hypothesis class consists of functions $f_{\boldsymbol{\theta}}(x) = wx + b$
- parameters: $\boldsymbol{\theta} = (b, w)^T \in \mathbb{R}^2$
- consider $\boldsymbol{\theta}^{(1)} = (150, 0.5)^T$ and $\boldsymbol{\theta}^{(2)} = (84, 1.7)^T$
- which one is better? in what sense it is better?

- consider an arbitrary training point $\boldsymbol{x}^{(i)}$, $y^{(i)}$
- consider parameters $\boldsymbol{\theta} = (b, \overbrace{w_1, \ldots, w_D}^{\boldsymbol{w}^T})^T$
- he **predicted target** is $\hat{y}^{(i)} =: f_{\boldsymbol{\theta}}(\boldsymbol{x}^{(i)}) = \boldsymbol{w}^T \boldsymbol{x}^{(i)} + b$
- the **squared error** (ℓ_2 **error**) between **predicted** and **true** target is

$$\Big(\underbrace{\hat{y}^{(i)} - y^{(i)}}_{\text{prediction error } e^{(i)}} \Big)^2 = \Big(f_{\boldsymbol{\theta}}(\boldsymbol{x}^{(i)}) - y^{(i)} \Big)^2 = \Big(\boldsymbol{w}^T \boldsymbol{x}^{(i)} + b - y^{(i)} \Big)^2$$


190
185
180
175
170
165
160
48
50
52
54
56
58
$y^{(i)}$
$\hat{y}^{(i)}$
$x^{(i)}$
$\theta = (143, 0.6)^T$

- we want the squared error to be small on all training examples
- idea: average the squared error over the whole data set
- this yields the **least-squares loss function** (ℓ_2 **loss**):

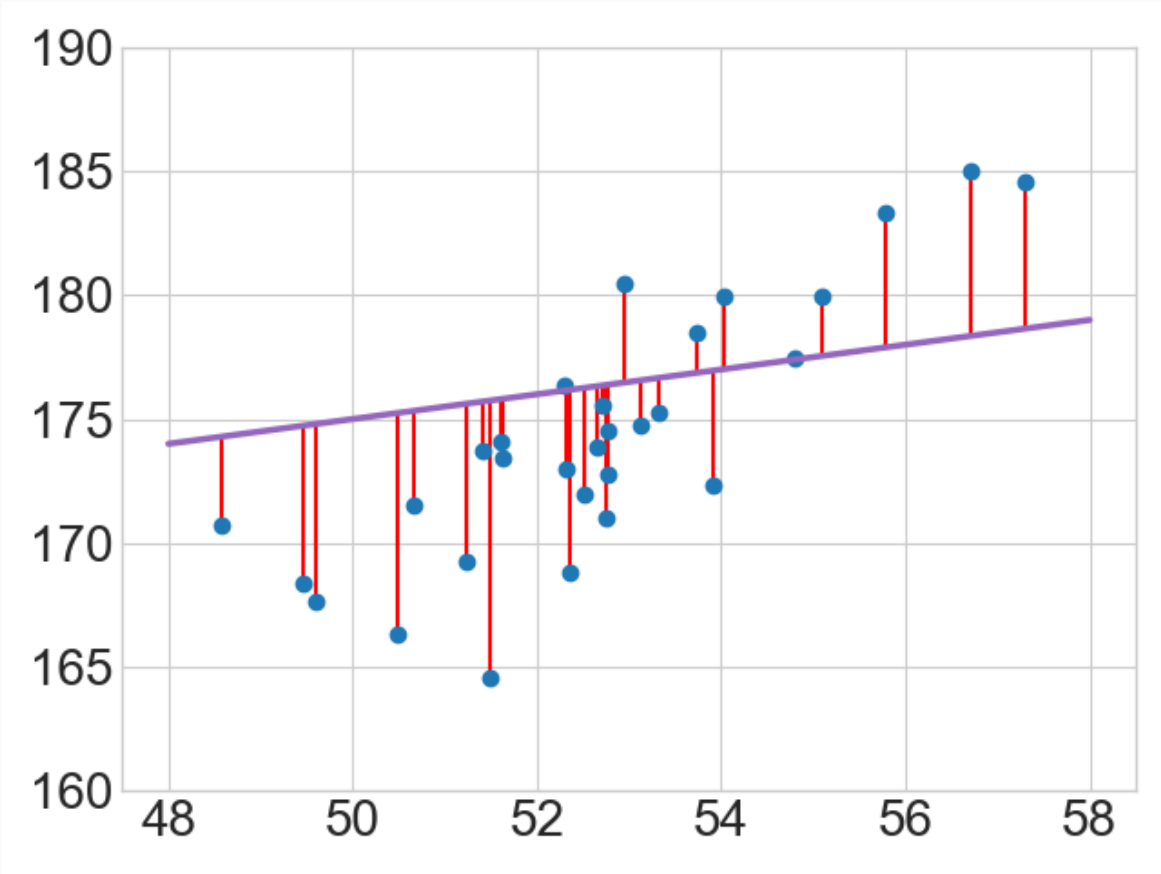
$$\begin{aligned}\mathcal{L}(\boldsymbol{\theta}) &= \frac{1}{N}\sum_{i=1}^{N}\left(\hat{y}^{(i)} - y^{(i)}\right)^2 \\ &= \frac{1}{N}\sum_{i=1}^{N}\left(f_{\boldsymbol{\theta}}(\boldsymbol{x}^{(i)}) - y^{(i)}\right)^2 \\ &= \frac{1}{N}\sum_{i=1}^{N}\left(\boldsymbol{w}^T\boldsymbol{x}^{(i)} + b - y^{(i)}\right)^2\end{aligned}$$

- a **loss function** (**objective**, **cost function**) is a notion of **fitness**
- the lower $\mathcal{L}(\boldsymbol{\theta})$, the better $\boldsymbol{\theta}$

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{N}\sum_{i=1}^{N}\left(f_{\boldsymbol{\theta}}(\boldsymbol{x}^{(i)}) - y^{(i)}\right)^2$$

$\boldsymbol{\theta}^{(1)} = (150, 0.5)^T$

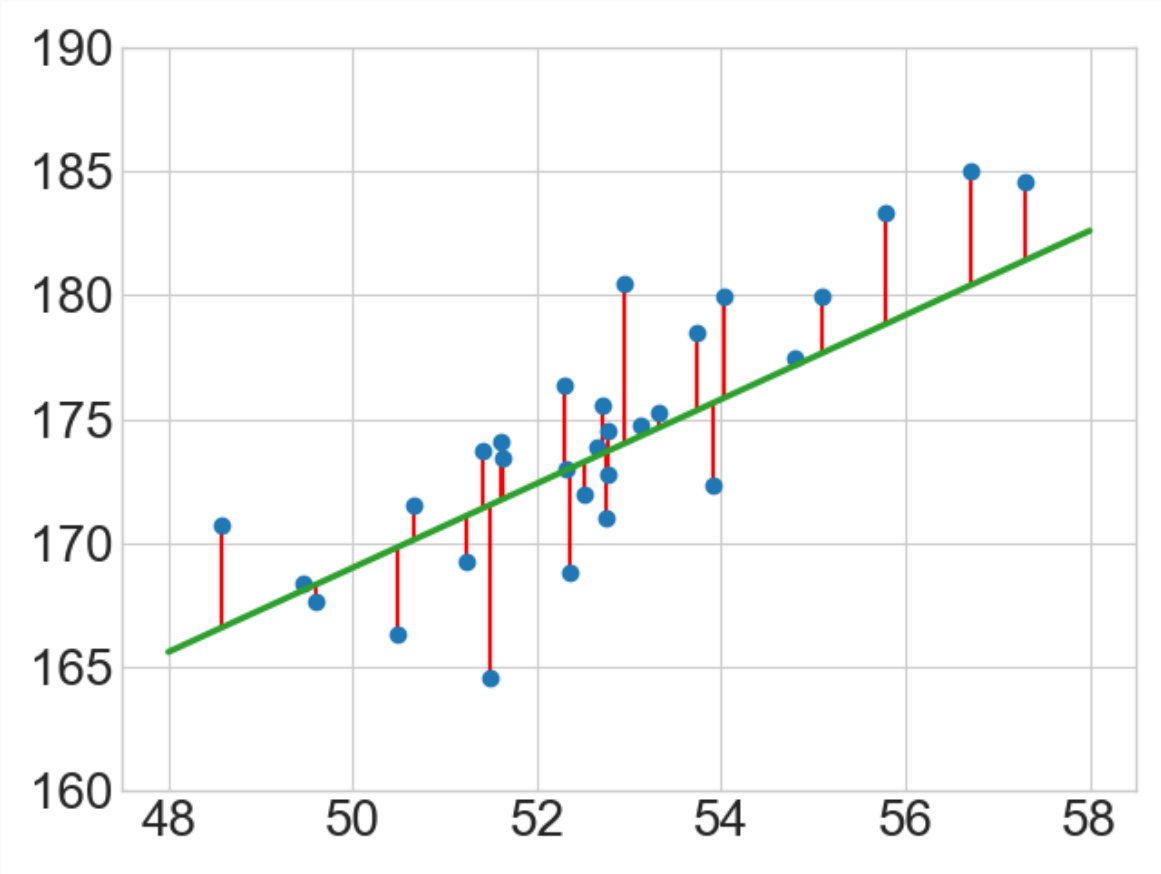
$\mathcal{L}(\boldsymbol{\theta}^{(1)}) = 22.67$



$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{N}\sum_{i=1}^{N}\left(f_{\boldsymbol{\theta}}(\boldsymbol{x}^{(i)}) - y^{(i)}\right)^2$$

$\boldsymbol{\theta}^{(2)} = (84, 1.7)^T$

$\mathcal{L}(\boldsymbol{\theta}^{(2)}) = 9.28$

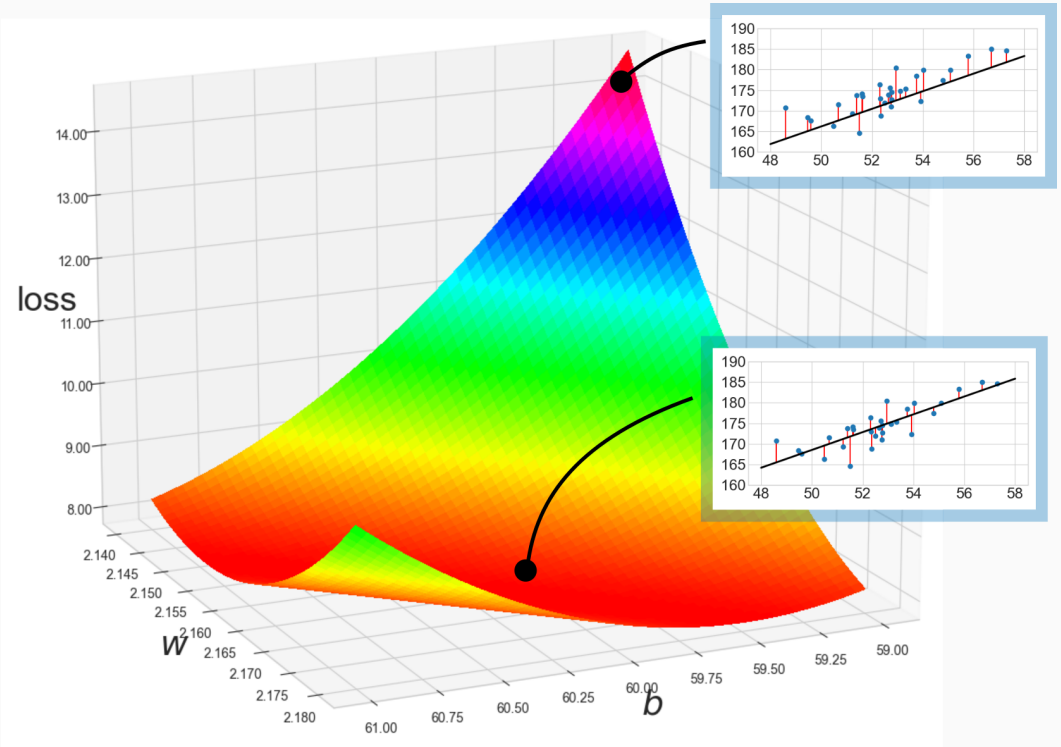


Loss Function Illustrated

Example

- ranging over all θ yields a "**loss landscape**"
- least squares: **quadratic function**
- **convex** ("bowl shaped")
- for convex functions, every **local minimum** is a **global minimum**

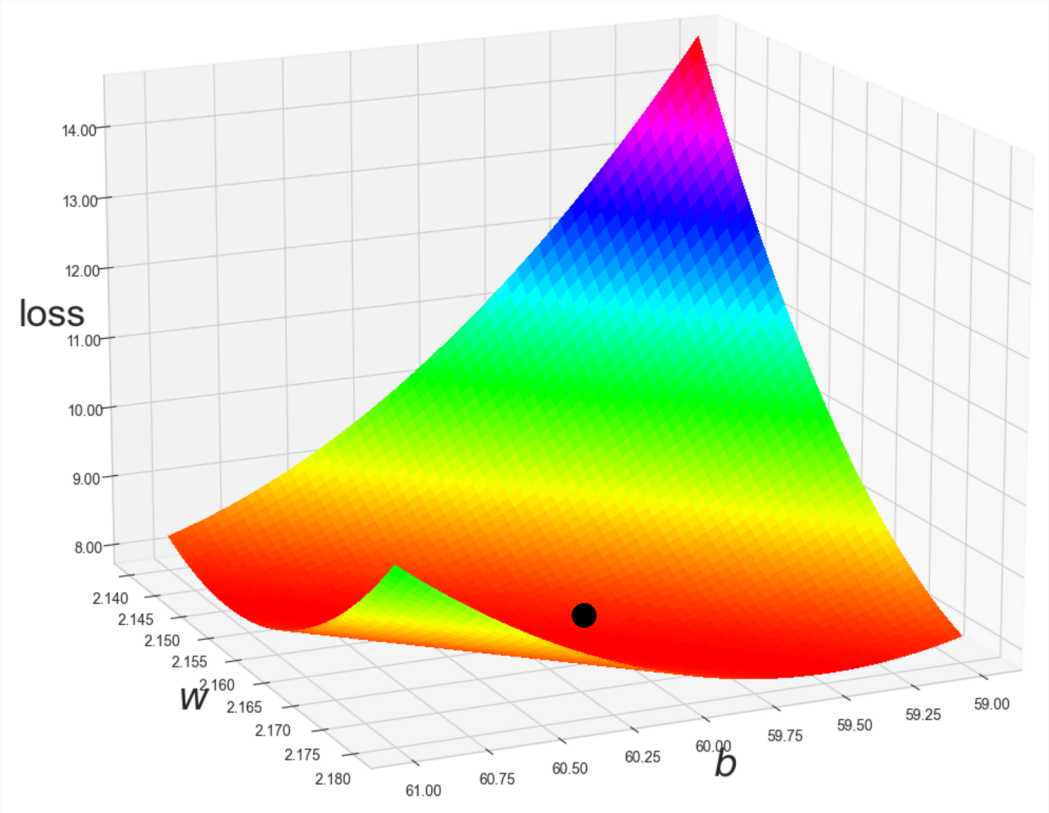
$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^{N} \left(f_\theta(\boldsymbol{x}^{(i)}) - y^{(i)} \right)^2$$



Least Squares — Analytic Solution

Minimum of $\mathcal{L}(\theta)$ (Quadratic Function)

The regression loss $\mathcal{L}(\theta)$ has (under regularity conditions) a **unique stationary point**, which is its **unique global minimum**.



Thus, we find the **least-squares solution** θ^* by setting $\nabla_\theta \mathcal{L} = 0$.

Absorbing the Bias

The model is an affine function

$$f_{\boldsymbol{\theta}}(\boldsymbol{x}) = \boldsymbol{w}^T \boldsymbol{x} + b = \sum_{i=1}^{D} w_i x_i + b$$

We can simplify notation a bit, by expanding the input vector by a **constant** (**"dummy feature"**):

$$\boldsymbol{x} := (1, x_1, x_2, \ldots, x_D)^T$$

$$f_{\boldsymbol{\theta}}(\boldsymbol{x}) = \sum_{i=1}^{D} w_i x_i + b = \sum_{i=1}^{D} \theta_i x_i + \theta_0 \, 1 = \boldsymbol{\theta}^T \boldsymbol{x},$$

where $\boldsymbol{\theta} = (b, w_1, \ldots, w_D)^T$.

Least-Squares in Matrix Form

- collect all $\boldsymbol{x}^{(i)}$ as rows in a $N \times (D+1)$ matrix:

$$\boldsymbol{X} = \begin{pmatrix} \boldsymbol{x}^{(1)\,T} \\ \boldsymbol{x}^{(2)\,T} \\ \vdots \\ \boldsymbol{x}^{(N)\,T} \end{pmatrix} = \begin{pmatrix} 1 & x_1^{(1)} & x_2^{(1)} & \dots & x_D^{(1)} \\ 1 & x_1^{(2)} & x_2^{(2)} & \dots & x_D^{(2)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_1^{(N)} & x_2^{(N)} & \dots & x_D^{(N)} \end{pmatrix}$$

- $\boldsymbol{X}$ is called the **design matrix**
- the vector of predictions is

$$\hat{\boldsymbol{y}} = \begin{pmatrix} \hat{y}^{(1)} \\ \hat{y}^{(2)} \\ \vdots \\ \hat{y}^{(N)} \end{pmatrix} = \begin{pmatrix} \boldsymbol{x}^{(1)\,T}\boldsymbol{\theta} \\ \boldsymbol{x}^{(2)\,T}\boldsymbol{\theta} \\ \vdots \\ \boldsymbol{x}^{(N)\,T}\boldsymbol{\theta} \end{pmatrix} = \begin{pmatrix} \boldsymbol{x}^{(1)\,T} \\ \boldsymbol{x}^{(2)\,T} \\ \vdots \\ \boldsymbol{x}^{(N)\,T} \end{pmatrix} \boldsymbol{\theta} = \boldsymbol{X}\boldsymbol{\theta}$$

Least-Squares in Matrix Form cont'd

- collect all targets $y^{(i)}$ in a vector:

$$\boldsymbol{y} = \begin{pmatrix} y^{(1)} \\ y^{(2)} \\ \vdots \\ y^{(N)} \end{pmatrix}$$

- the vector of **prediction errors** is given as

$$\boldsymbol{e} := \begin{pmatrix} \hat{y}^{(1)} - y^{(1)} \\ \hat{y}^{(2)} - y^{(2)} \\ \vdots \\ \hat{y}^{(N)} - y^{(N)} \end{pmatrix} = \hat{\boldsymbol{y}} - \boldsymbol{y} = \boldsymbol{X}\boldsymbol{\theta} - \boldsymbol{y}$$

Least-Squares in Matrix Form cont'd

- the least-squares objective can now be written

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{N}\sum_{i=1}^{N}\left(\hat{y}^{(i)} - y^{(i)}\right)^2 = \frac{1}{N}\sum_{i=1}^{N} e^{(i)} e^{(i)}$$
$$= \frac{1}{N}\boldsymbol{e}^T\boldsymbol{e}$$
$$= \frac{1}{N}\left(\boldsymbol{X}\boldsymbol{\theta} - \boldsymbol{y}\right)^T\left(\boldsymbol{X}\boldsymbol{\theta} - \boldsymbol{y}\right)$$

- **quadratic function** in $\boldsymbol{\theta}$
- the gradient can be shown to be

$$\nabla_{\boldsymbol{\theta}}\mathcal{L} = \frac{1}{N} 2\boldsymbol{X}^T\left(\boldsymbol{X}\boldsymbol{\theta} - \boldsymbol{y}\right)$$

- compare this with the 1-d case: $\frac{d}{\mathrm{d}\theta}(x\theta - y)^2 = 2x(x\theta - y)$

Closed Form Solution

Setting the gradient to zero yields the minimum:

$$\nabla_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}) = \frac{1}{N} 2 \boldsymbol{X}^T (\boldsymbol{X}\boldsymbol{\theta} - \boldsymbol{y}) \stackrel{!}{=} 0$$
$$\boldsymbol{X}^T (\boldsymbol{X}\boldsymbol{\theta} - \boldsymbol{y}) = 0$$

Hence, we have to solve a linear system of equations:

$$\boldsymbol{X}^T \boldsymbol{X} \boldsymbol{\theta}^* = \boldsymbol{X}^T \boldsymbol{y}$$
$$\boldsymbol{\theta}^* = \underbrace{(\boldsymbol{X}^T \boldsymbol{X})^{-1} \boldsymbol{X}^T}_{\text{Moore-Penrose Inverse}} \boldsymbol{y}$$

$(\boldsymbol{X}^T \boldsymbol{X})^{-1}$ exist, if the columns of $\boldsymbol{X}$ are **linearly independent** i.e., when there are no "redundant features". These is also a regularity condition ensuring that the global minimum is unique.

Closed Form Solution

Example

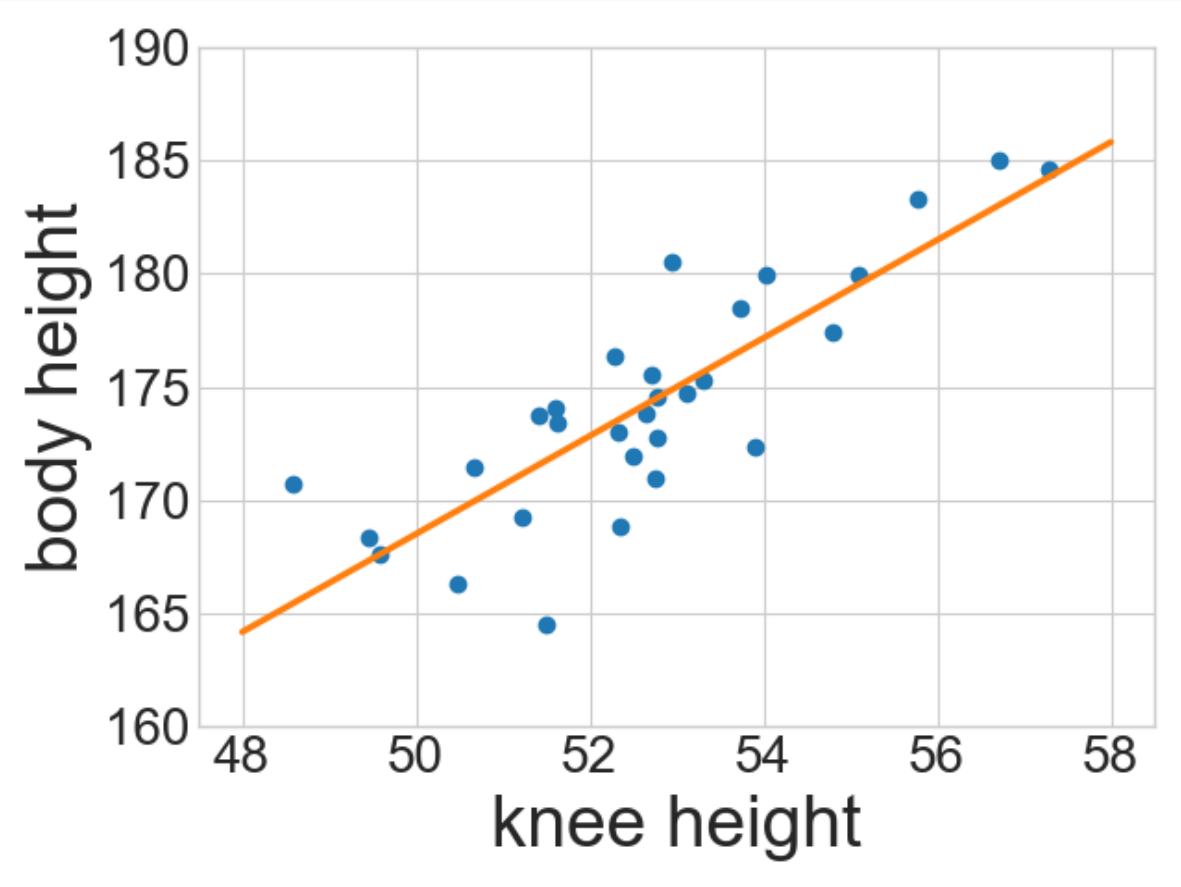
n	knee height	height
1	56.7	185.0
2	54.8	177.4
3	52.7	175.5
4	51.5	164.6
5	51.6	173.4
6	52.3	173.0
7	52.8	174.5
8	50.7	171.5
9	53.1	174.8
10	53.9	172.3
11	49.6	167.6
12	55.1	180.0
13	54.0	179.9
14	57.3	184.6
15	53.3	175.3

n	knee height	height
16	52.8	172.7
17	52.5	171.9
18	50.5	166.3
19	52.9	180.5
20	52.4	168.8
21	49.5	168.4
22	52.7	173.8
23	55.8	183.3
24	52.3	176.3
25	51.2	169.3
26	52.8	171.0
27	53.7	178.5
28	51.6	174.1
29	48.6	170.7
30	51.4	173.7

$$\boldsymbol{X} = \underbrace{\begin{pmatrix} 1 & 56.7 \\ 1 & 54.8 \\ 1 & 52.7 \\ \vdots & \vdots \end{pmatrix}}_{30\times 2} \quad \boldsymbol{y} = \underbrace{\begin{pmatrix} 185 \\ 177.4 \\ 175.5 \\ \vdots \end{pmatrix}}_{30\times 1}$$

$$\boldsymbol{X}^T\boldsymbol{X} = \begin{pmatrix} 30 & 1580.1 \\ 1580.1 & 83339.25 \end{pmatrix} \qquad (\boldsymbol{X}^T\boldsymbol{X})^{-1} = \begin{pmatrix} 24.076 & -0.456 \\ -0.456 & 0.009 \end{pmatrix}$$

$$\boldsymbol{X}^T\boldsymbol{y} = \begin{pmatrix} 5228.7 \\ 275645.13 \end{pmatrix} \qquad \boldsymbol{\theta}^* = (\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T\boldsymbol{y} = \begin{pmatrix} 60.4 \\ 2.16 \end{pmatrix}$$



Optimal in least-squares sense: $\boldsymbol{\theta}^* = \begin{pmatrix} 60.4 \\ 2.16 \end{pmatrix}$ $\mathcal{L}(\boldsymbol{\theta}^*) = 7.95$

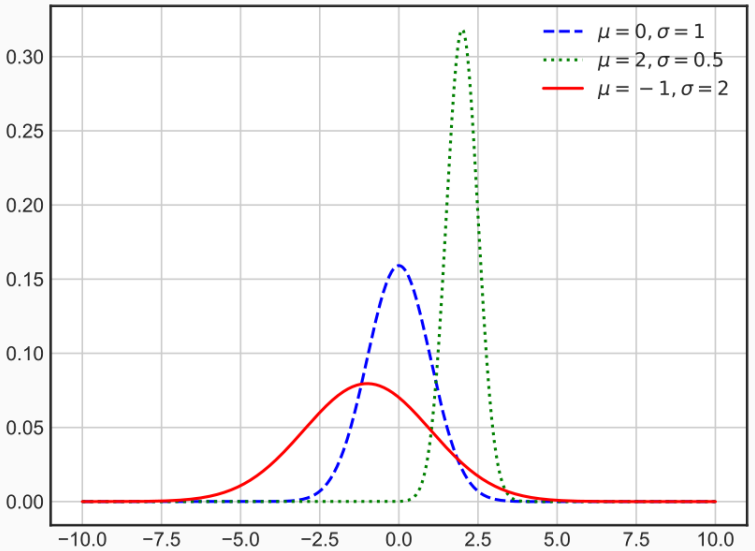
Connection to Maximum Likelihood

- least-squares has an interesting connection to **Gaussian density estimation** and the **maximum likelihood principle**
- we will first review Gaussian density estimation, an **unsupervised** technique, and connect it then to linear regression, a **supervised** technique

Consider the univariate **Gaussian distribution** with density

$$p(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

with parameters **mean** μ and **standard deviation** $\sigma > 0$, which might be written as parameter vector $\boldsymbol{\theta} = (\mu, \sigma)^T$.



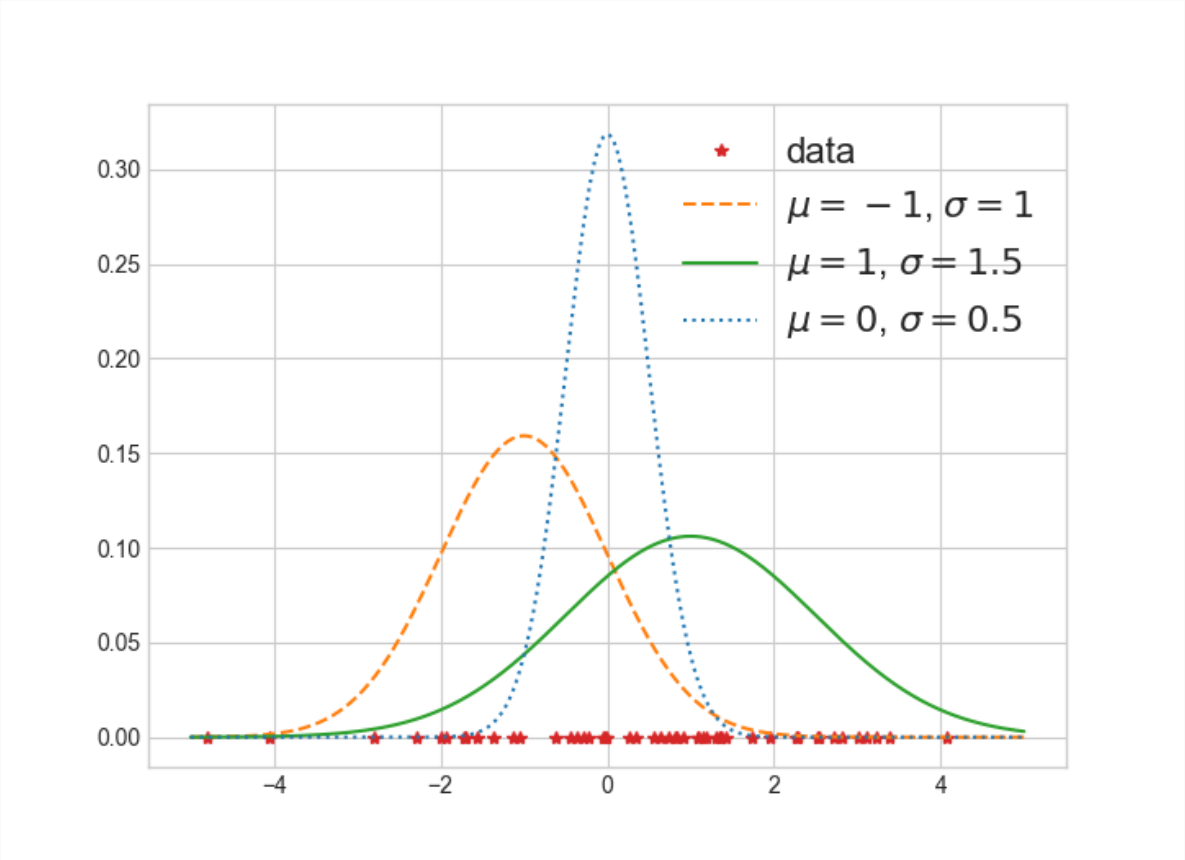


Carl Friedrich Gauss 1777–1855

Image: wikipedia

How to fit a Gaussian?

Say we have observed data $\mathcal{D} = \{x^{(1)}, \ldots, x^{(N)}\}$? Which $\boldsymbol{\theta} = (\mu, \sigma)^T$ explain this data best? What is a good objective function for density estimation?



Note that there are only inputs now – density estimation is **unsupervised**.

When $\mathcal{D} = \{x^{(1)}, x^{(2)}, \ldots, x^{(N)}\}$ are drawn **independently and identically distributed (i.i.d.)**, the probability density of the whole dataset $\mathcal{D}$ is

$$p(\mathcal{D}; \boldsymbol{\theta}) = \prod_{i=1}^{N} p(x^{(i)}; \boldsymbol{\theta})$$

Note: Generally, multiple random variables $X_1, X_2, \ldots, X_N$ are **statistically independent** if their distribution factorizes. Hence, the probability density $p(\mathcal{D}; \boldsymbol{\theta})$ factorizes into **sample-wise** densities $p(x^{(i)}; \boldsymbol{\theta})$ due to the i.i.d. assumption.

$p(\mathcal{D}; \boldsymbol{\theta})$ depends on the dataset $\mathcal{D}$ and the parameters $\boldsymbol{\theta}$.

- when seen as function of data: probability density of $\mathcal{D}$
- when seen as function of parameters: **likelihood** of $\boldsymbol{\theta}$

Maximum Likelihood Estimator

Parameters $\boldsymbol{\theta}^*$ which maximize the likelihood

$$\boldsymbol{\theta}^* := \arg\max_{\boldsymbol{\theta}} \; p(\mathcal{D}; \boldsymbol{\theta}) = \arg\max_{\boldsymbol{\theta}} \; \prod_{i=1}^{N} p(x^{(i)}; \boldsymbol{\theta})$$

Generally, for any parametric model (not only Gaussians), maximum likelihood is a **consistent estimator,** i.e. for $N \to \infty$ it recovers the distribution closest to the true data distribution.

Instead of likelihood, we can maximize the **log-likelihood**:

$$\log p(\mathcal{D}; \boldsymbol{\theta}) := \log \prod_{i=1}^{N} p(x^{(i)}; \boldsymbol{\theta}) = \sum_{i=1}^{N} \log p(x^{(i)}; \boldsymbol{\theta})$$

Since the log is a **strictly increasing** function, the log-likelihood has exactly the same maxima as the likelihood, i.e. maximizing log-likelihood is equivalent to maximizing likelihood.

Optimizing the log-likelihood is usually easier and numerically more stable than optimizing likelihood (product of many small factors).

Maximum Likelihood for Gaussians

- Gaussian likelihood: $p(\mathcal{D}; \mu, \sigma) = \prod_{i=1}^{N} \frac{1}{\sqrt{2\pi}\,\sigma} e^{-\frac{1}{2}\left(\frac{x^{(i)}-\mu}{\sigma}\right)^2}$
- thus, the log-likelihood is

$$\log p(\mathcal{D}; \mu, \sigma) = \sum_{i=1}^{N} -\log(\sqrt{2\pi}\,\sigma) - \frac{1}{2}\left(\frac{x^{(i)}-\mu}{\sigma}\right)^2$$

- **(negative) quadratic function** in μ
- deriving after μ delivers:

$$\frac{\partial \log p}{\partial \mu} = \sum_{i=1}^{N} -\frac{1}{2}\,2\left(\frac{x^{(i)}-\mu}{\sigma}\right)\left(-\frac{1}{\sigma}\right) = \sum_{i=1}^{N} \frac{x^{(i)}-\mu}{\sigma^2}$$

- at maximum, $\frac{\partial \log p(\mathcal{D};\mu,\sigma)}{\partial \mu} \stackrel{!}{=} 0$

Maximum Likelihood for Gaussians cont'd

$$\frac{\partial \log p(\mathcal{D}; \mu, \sigma)}{\partial \mu} = \sum_{i=1}^{N} \frac{x^{(i)} - \mu}{\sigma^2} \stackrel{!}{=} 0$$

Hence,

$$\sum_{i=1}^{N} x^{(i)} = \sum_{i=1}^{N} \mu^* = N\mu^* \quad \Rightarrow \quad \mu^* = \frac{1}{N} \sum_{i=1}^{N} x^{(i)}$$

- Maximum likelihood μ^* is simply the **average** over data
- Solution independent of σ
- Thus, we can now fix μ^* and maximize w.r.t. σ

Maximum Likelihood for Gaussians: σ^*

$$\log p(\mathcal{D}; \mu^*, \sigma) = \sum_{i=1}^{N} -\log(\sqrt{2\pi}\,\sigma) - \frac{1}{2}\left(\frac{x^{(i)} - \mu^*}{\sigma}\right)^2$$

Deriving after σ:

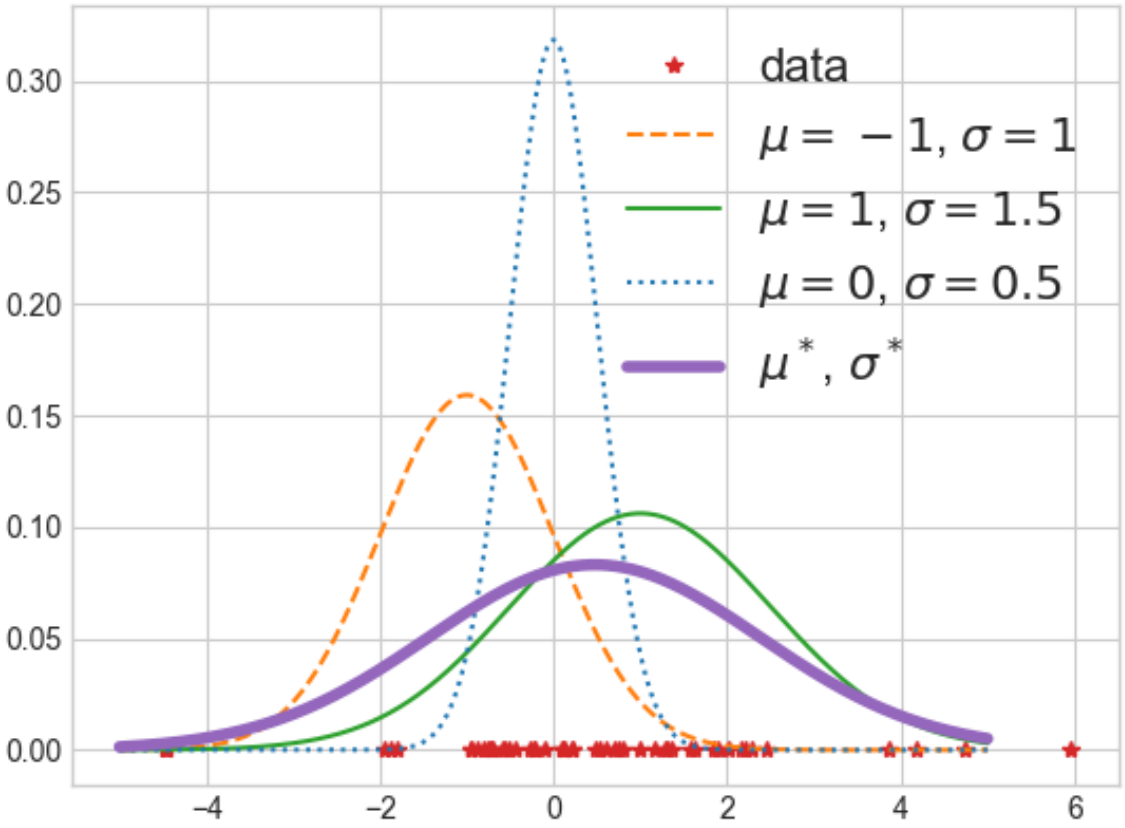
$$\frac{\partial \log p}{\partial \sigma} = \sum_{i=1}^{N} -\frac{\sqrt{2\pi}}{\sqrt{2\pi}\,\sigma} - \frac{1}{2}\,2\left(\frac{x^{(i)} - \mu^*}{\sigma}\right)(-1)\left(\frac{x^{(i)} - \mu^*}{\sigma^2}\right)$$

$$= \sum_{i=1}^{N} -\frac{1}{\sigma} + \left(\frac{(x^{(i)} - \mu^*)^2}{\sigma^3}\right) \stackrel{!}{=} 0$$

Hence, we get the **empirical standard deviation** as ML solution:

$$\sum_{i=1}^{N} \sigma^2 = \sum_{i=1}^{N} (x^{(i)} - \mu^*)^2 \quad \Rightarrow \quad \sigma^* = \sqrt{\frac{1}{N}\sum_{i=1}^{N} (x^{(i)} - \mu^*)^2}$$

The maximum likelihood solution



Connection to Linear Regression

Connection to Linear Regression

- Assume now a **regression problem** with data

$$\mathcal{D} = \left\{ (\boldsymbol{x}^{(1)}, y^{(1)}), (\boldsymbol{x}^{(2)}, y^{(2)}), \ldots, (\boldsymbol{x}^{(N)}, y^{(N)}) \right\}$$

- We model now **the output** $y^{(i)}$ as univariate Gaussian with
  - **fixed** standard deviation σ,
  - but whose **mean depends on** $\boldsymbol{x}^{(i)}$:

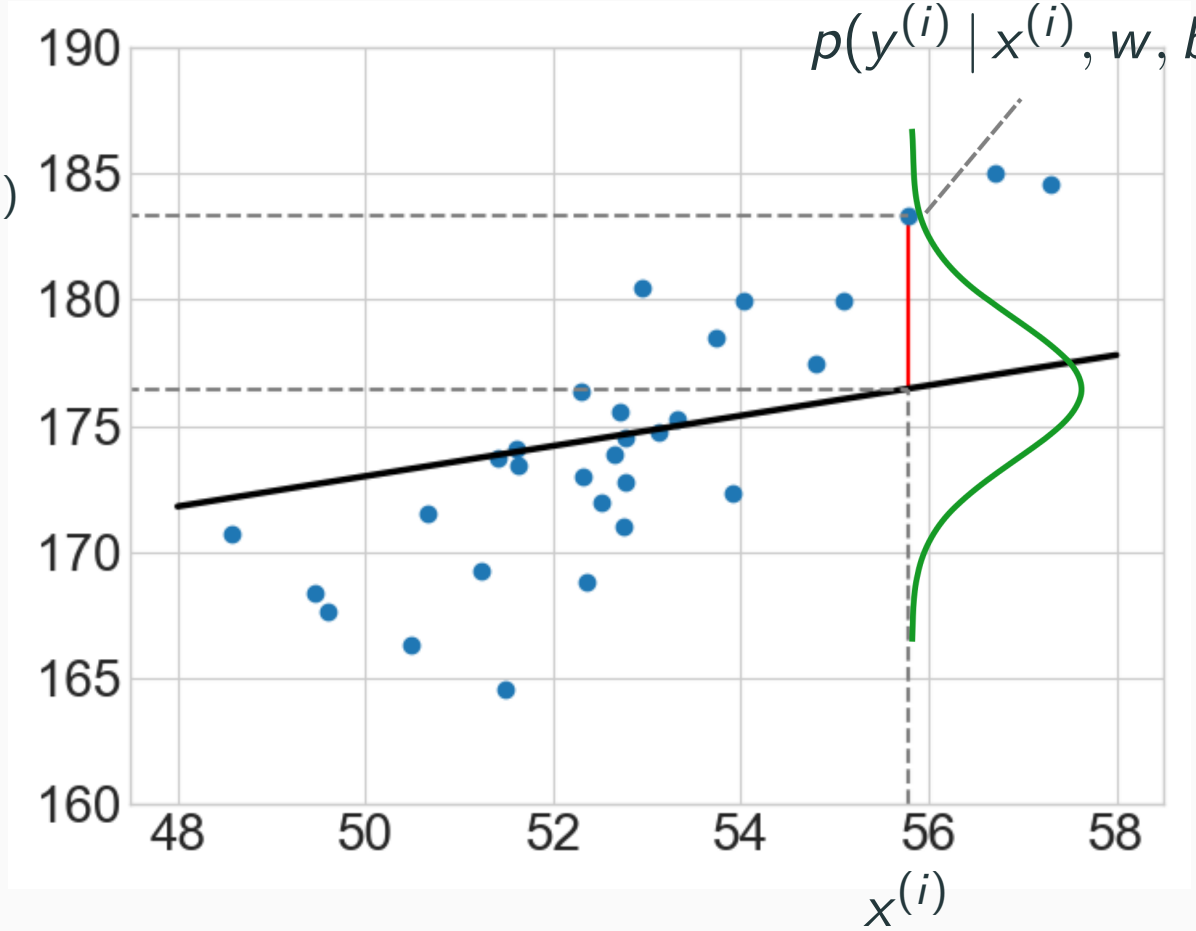
$$\mu^{(i)} = \boldsymbol{w}^T \boldsymbol{x}^{(i)} + b$$

- Thus, $y^{(i)}$ is drawn from a **conditional Gaussian**:

$$p(y^{(i)} \mid \boldsymbol{x}^{(i)}, \boldsymbol{w}, b, \sigma) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{-\frac{1}{2}\left(\frac{y^{(i)} - \boldsymbol{w}^T \boldsymbol{x}^{(i)} - b}{\sigma}\right)^2}$$

- Equivalently, the prediction error $e^{(i)} = y^{(i)} - \boldsymbol{w}^T \boldsymbol{x}^{(i)} - b$ is assumed to be Gaussian with $\mu = 0$, and fixed σ

$$\mu^{(i)} = wx^{(i)} + b$$


$p(y^{(i)} \mid x^{(i)}, w, b, \sigma)$
$y^{(i)}$
190
185
180
175
170
165
160
48
50
52
54
56
58
$x^{(i)}$

Maximum Likelihood for Linear Regression

- let $\boldsymbol{y} = (y^{(1)}, y^{(2)}, \ldots, y^{(N)})^T$ be the target vector
- $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(N)}$ are fixed, including a "dummy feature" for the bias
- $\boldsymbol{\theta} = (b, w_1, \ldots, w_D)^T$ are model parameters to be learned
- thus, we can simply write $\mu^{(i)} = \boldsymbol{\theta}^T \boldsymbol{x}^{(i)}$
- standard deviation σ is fixed
- we want to maximize the log-likelihood:

$$\boldsymbol{\theta}^* = \arg\max_{\boldsymbol{\theta}} \log p(\boldsymbol{y} \mid \boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(N)}, \boldsymbol{\theta}, \sigma)$$

Maximum Likelihood for Linear Regression

$$
\begin{aligned}
&\log p(\boldsymbol{y} \mid \boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(D)}, \boldsymbol{\theta}, \sigma) \\
&\quad = \log \prod_{i=1}^{N} p(y^{(i)} \mid \boldsymbol{x}^{(i)}, \boldsymbol{\theta}, \sigma) \\
&\quad = \sum_{i=1}^{N} \log \left(\frac{1}{\sqrt{2\pi}\,\sigma} e^{-\frac{1}{2}\left(\frac{y^{(i)} - \boldsymbol{\theta}^T \boldsymbol{x}^{(i)}}{\sigma}\right)^2} \right) \\
&\quad = \sum_{i=1}^{N} \overbrace{-\log \sqrt{2\pi}\,\sigma}^{\text{const.}} - \overbrace{\frac{1}{2\sigma^2}}^{\textit{scaling}} \left(y^{(i)} - \boldsymbol{\theta}^T \boldsymbol{x}^{(i)} \right)^2
\end{aligned}
$$

Compare this with the least squares objective:

$$
\overbrace{\frac{1}{N}}^{\textit{scaling}} \sum_{i=1}^{N} \left(y^{(i)} - \boldsymbol{\theta}^T \boldsymbol{x}^{(i)} \right)^2
$$

Maximum Likelihood for Linear Regression

Clearly,

$$\max_{\boldsymbol{\theta}} \quad -\frac{1}{2\sigma^2} \sum_{i=1}^{N} \left(y^{(i)} - \boldsymbol{\theta}^T \boldsymbol{x}^{(i)} \right)^2$$

is equivalent to

$$\min_{\boldsymbol{\theta}} \quad \frac{1}{N} \sum_{i=1}^{N} \left(y^{(i)} - \boldsymbol{\theta}^T \boldsymbol{x}^{(i)} \right)^2$$

Thus, minimizing least squares is equivalent to maximum likelihood under Gaussian output assumption, with fixed σ!

Equivalence to Probabilistic Model

Why does this matter?

- interpretability and understanding methods in different ways
- probabilistic analysis and justification
- making the implicit model assumptions explicit
- generalizing the model by playing with probabilistic assumptions
- generally, many apparently non-probabilistic machine learning algorithms can be explained with probabilistic arguments